Exploring Directional Sparsity and Cross-Modal Contrast-Consensus: MERGE-Net for HSI–SAR/LiDAR Joint Classification

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***Abstract*—Multimodal satellite imagery can provide more comprehensive information, which shows a new perspective for observing Earth's surface. Particularly, the joint use of hyperspectral image (HSI), synthetic aperture radar (SAR), or light detection and ranging (LiDAR) has gained widespread attention due to the concurrent acquisition of spectral–spatial information, structural information, or ground elevation. However, most existing works do not explicitly learn effective intramodal and intermodal mutual information. This induces information redundancy and curtails feature diversity, stalling performance advances across heterogeneous modalities. To address the above challenge, a unified mutual learning with geometric equilibrium network (MERGE-Net) is proposed for HSI–SAR/LiDAR collaborative classification. Concretely, a modality-specific structured sparse modeling module is first designed for intramodal mutual learning, which selectively samples structurally representative neighbors using an anisotropic, axial-favoring sampling prior, thereby suppressing redundancy while enhancing discriminative spectral–spatial cues. Then, a continuous contrast-consensus integration (C3I) mechanism is put forward for intermodal mutual learning, where complementary information and consistent semantics are iteratively propagated to align heterogeneous modalities at multiple semantic depths. Finally, a geometric equilibrium equiangular tight frame head is devised to suppress minority collapse and stabilize decision boundaries in label-scarce remote sensing data. In this way, MERGE-Net can simultaneously explore intramodal and intermodal mutual relationships while enhancing discriminative and balanced class separation, offering a new pathway to multimodal classification. Experiments and analysis on three benchmark datasets implicate the effectiveness of the proposed approach compared with state-of-the-art (SOTA) methods. The code will be released at: https://github.com/jqyang22/MERGE-Net upon publication.**

***Index Terms*—Hyperspectral image (HSI)–synthetic aperture radar (SAR)/light detection and ranging (LiDAR) imagery, intramodal and intermodal modeling, multimodal collaborative classification, unified mutual learning.**

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I. Introduction

As an emerging technological product, hyperspectral remote sensing imagery has high spectral resolution and numerous bands, which allow the Earth's surface to be probed on a microscopic scale [1], [2], [3]. However, hyperspectral image (HSI) often struggles to separate land covers with near-identical spectra because it lacks explicit structural and elevation cues [4]. Synthetic aperture radar (SAR) and light detection and ranging (LiDAR) complement HSI [5], [6]. SAR contributes geometry- and texture-sensitive backscatter with day–night, cloud-robust acquisition, while LiDAR provides accurate height information. This complementarity has driven a shift from single- to multimodal Earth observation and enabled advances in applications such as crop monitoring [7], urban analysis [8], and water resources [9]. In the face of such rich and diverse data, how to design effective and appropriate approaches for multisensor image joint classification has become a key challenge [10], [11].

Deep models dominate multisource classification [12], [13]. Transformer-based approaches leverage long-range self-attention and have been tailored for cross-modal learning via hierarchical encoders, global–local designs, mutual-benefit attention, and spectral–spatial morphological attention [14], [15]. Zhao et al. [16] presented a hierarchical Transformer encoder to capture cross-modal features. In [17], a novel global–local Transformer network (GLTNet) was put forward for simultaneous modeling of short- and long-range dependencies. Aiming at nonlocal multimodal feature fusion, Wang and Tan [18] proposed the mutually beneficial Transformer. Moreover, a spectral–spatial morphological attention Transformer [19] was introduced to improve the interplay between structural properties and shape information. Although the aforementioned studies achieve promising performance, they fail to explicitly compute the spectral–spatial signals and ignore the native correlation of space and spectrum in remote sensing imagery [14].

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As another typical structure, convolutional neural network (CNN) has also received a wide range of applications to improve the interpretation performance. Thanks to their inherently 3-D operations, 3-D CNNs align naturally with spatial–spectral joint modeling of remote sensing images. Pande and Banerjee [20] proposed a coupled self-looping convolution network with 3-D convolution to reconstruct cross-modal information. Wei et al. [21] designed a quaternion CNN to encode internal and external relationships. To capture the variability of information, Wang et al. [22] proposed an uncertainty-aware dynamic fusion network. Furthermore, Wang et al. [23] devised a nearest neighbor-based contrastive learning to explore discriminative multisource data expression. The above approaches can better extract spectral–spatial cooperative information, whereas intramodal mutual feature extraction is not sufficient. Moreover, the well-documented inefficiency of 3-D convolutions has yet to be alleviated, posing significant challenges for large-scale applications [24], [25]. Although approaches such as dilated convolution, factorized/asymmetric 3-D convolution, and deformable convolution can reduce computational cost or enlarge the receptive field, their sampling patterns are either isotropic and modality-agnostic or learned purely from data, making them potentially unstable under the label-scarce conditions typical of remote sensing [26], [27]. Critically, none explicitly incorporates the directional structure of spectral–spatial HSI or spatial–structural SAR/LiDAR data as a sampling prior.

To further achieve cross-modal coordination, most existing studies focus on the multimodal feature fusion process [28], [29]. These strategies fall into three groups: early fusion combines modalities at the input or shallow layers; intermediate fusion enables feature-level interaction inside the network; and late fusion aggregates scores at the output layer. Hong et al. [30] explored early fusion with a concatenation operator. In [31], concatenation is utilized to fuse intermediate features generated from multiple branches. To interact with multisource information, Yang et al. [32] designed a heterogeneous encoder for intermediate fusion. Lin et al. [33] proposed a dynamic cross-modal feature interaction network (DCMNet) with addition calculation to aggregate HSI and LiDAR data for late fusion. Nevertheless, these methods attenuate intermodal mutual learning, including the extraction of differential cues and the reinforcement of consensus representations, thereby limiting fusion performance [34], [35].

In addition, due to reliance on expert prior knowledge, field survey bottlenecks, and spectral mixing issue, annotating HSI is exceptionally difficult. In most datasets, the labeled proportion is typically below 5%, resulting in a severe scarcity of annotated samples [36], [37]. Nevertheless, the prevailing practice is to use the linear softmax and cross-entropy head [29], sometimes augmented by resample/reweighting structures (focal [38]) or metric/prototype objectives (contrastive learning [39]) for the decision classifier. To a large extent, these are post hoc fixes. The geometry of class prototypes is left unconstrained, so interclass margins are uneven. Under severe imbalance, networks thus remain vulnerable to neural/minority collapse and biased decision boundaries [40]. As depth grows, the final head often becomes the bottleneck that is decisive for how features are projected and separated. Without explicit structural constraints, feature clusters drift into unstable and messy arrangements [41], [42].

In light of the above analysis, existing approaches face three principal issues: weak mutual learning within each modality, inadequate mutual learning across modalities, and limited classifier discriminability [43]. To address the above obstacles, a unified mutual learning with geometric equilibrium network (MERGE-Net) is proposed. It comprises three components: a direction-oriented sparse modeling (DOSM) module, a continuous contrast-consensus integration (C3I) mechanism, and a geometric equilibrium equiangular tight frame (GEETF) head.

The DOSM module uses multigroup, direction-oriented sparse 3-D convolutions. As illustrated in Fig. 1, vanilla 3-D convolution employs fixed, isotropic $3 \times 3 \times 3$ kernels that densely sample all 27 voxels, blurring boundaries, adding redundancy, increasing overfitting risk, and incurring high computational load. In contrast, DOSM applies a fixed, anisotropic sparse support and performs selective proximal sampling, retaining structurally salient voxels. The support is a modality-specific prior selected by channel count, not a per-location learned direction; adaptivity enters only through the convolutional weights defined over it. This structured sparsity preserves informative spectral–spatial cues and yields a low-cost effective kernel suited to heterogeneous modalities and large-scale scenes. The markedly lower voxel cuts run-time without sacrificing discriminative ability [44], [45].

Fig. 2 contrasts three common fusion operators with our C3I. Addition is parameter-free but averages away modality-specific cues and can be dominated by the stronger stream. Concatenation preserves scales in separate channels yet inflates dimensionality and relies on a 1×1 convolution that may overfit and remix channels without enforcing cross-scale alignment. Attention offers dynamic reweighting and noise suppression but adds parameters and often entangles discrepant and common features, leaving contrasts under-exploited. C3I alternates a subtraction branch, which isolates cross-modal differences, with an absolute-value summation branch, which reinforces aligned consensus. In this way, discriminative contrasts are retained while shared semantics are consolidated at modest cost, offering a balanced expressiveness-efficiency tradeoff for layerwise multimodal interaction.

The main contributions of the proposed work can be summarized as follows.

1) MERGE-Net illustrated in Fig. 3 is proposed for HSI–SAR/LiDAR joint classification. It unifies intramodal and intermodal mutual learning, allowing differences and commonalities to be refined concurrently throughout the depth of the model. With competitive inference latency, MERGE-Net leads to more distinctive and resilient representation learning.
2) A modality-specific structured sparse modeling module is introduced, in which a small set of structurally salient voxels is sampled using a fixed, anisotropic support whose geometry is conditioned on the channel count of each modality. The sampling support is a directional (axial-favoring) prior rather than a

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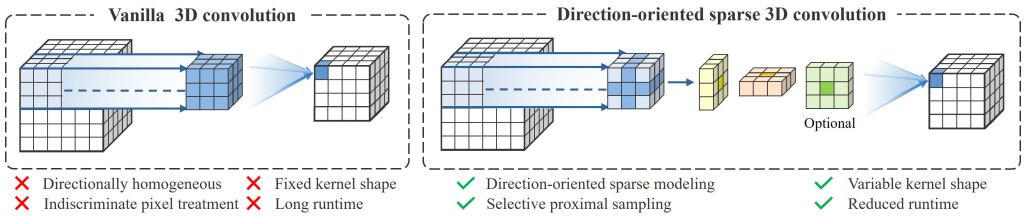


Fig. 1. Comparison of vanilla 3-D convolution and the proposed structured anisotropic sparse 3-D convolution.

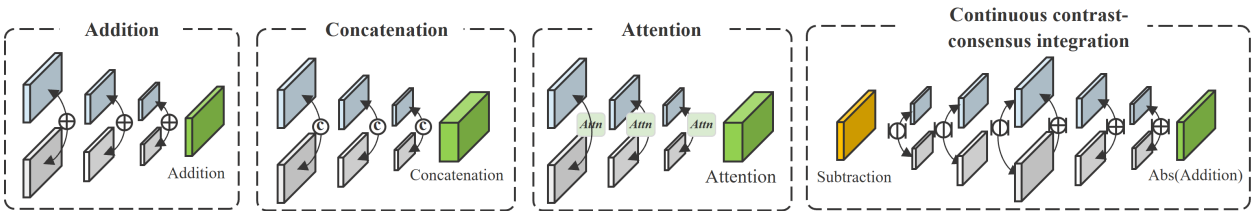


Fig. 2. Popular multimodal feature fusion methods and C3I.

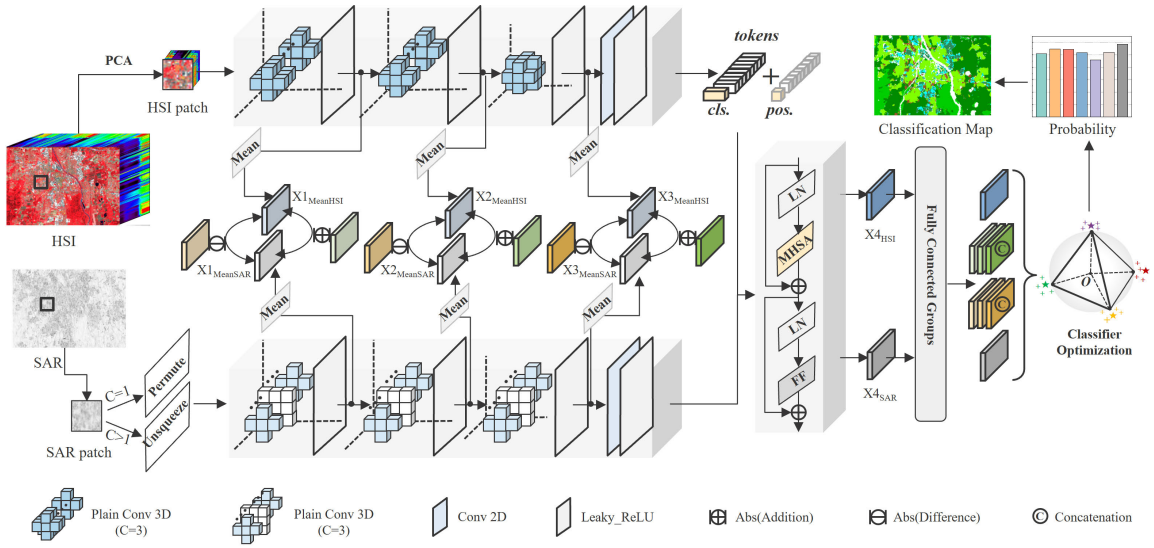


Fig. 3. Architecture of MERGE-Net (illustrated with HSI and SAR as inputs).

learned per-location orientation; learning occurs in the convolutional weights defined over this support. This module preserves anisotropic detail, suppresses redundant responses, and reduces time cost compared with vanilla dense $3 \times 3 \times 3$ kernels.

3) We design a C3I structure with residual subtraction and absolute-value addition. The subtraction branch highlights cross-modal discrepancies (contrast), while the addition branch progressively consolidates the aligned agreement (consensus), leading to discriminative and noise-robust representations.
4) The GEETF head is defined to maximize interclass margins, mitigate minority-class collapse in imbalanced data, and yield a decision layer that complements the preceding feature extraction.

The remainder of this article is organized as follows. Section II introduces the proposed approach in detail. Section III reports experiments and discussions. Conclusion and future work are presented in Section IV.

II. Methodology

A. Preprocessing

HSI typically covers hundreds of bands from the ultraviolet to the near-infrared, supplying abundant spectral information but also imposing heavy computational demands and the Hughes phenomenon [46], [47]. Therefore, the principal component analysis (PCA) algorithm [48] is utilized for compressing the data cube while keeping the information more beneficial.

Let $X_h \in \mathbb{R}^{H\times W\times B_h}$ and $X_{s/l} \in \mathbb{R}^{H\times W\times B_{s/l}}$ be the HSI and SAR/LiDAR imagery after normalization, where H, W, B_h, and $B_{s/l}$ denote the height, width, bands in HSI, and bands in SAR/LiDAR data, respectively. After PCA, HSI and SAR/LiDAR imagery are projected to $X_h^{pca} \in \mathbb{R}^{H\times W\times C_h}$ and $X_{s/l}^{pca} \in \mathbb{R}^{H\times W\times C_{s/l}}$, where C_h and $C_{s/l}$ are preserved principal components for the corresponding modality.

For each center pixel, a neighborhood patch is extracted from each modality. In the HSI/SAR cube, this patch forms a

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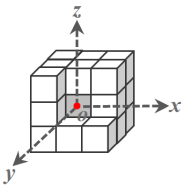


Fig. 4. HSI cube coordinate system.

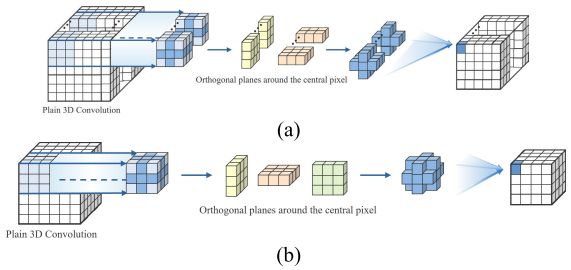


Fig. 5. Two variants of the direction-oriented sparse convolution for different channel numbers. (a) Structure when the number of channels > 3. (b) Structure when the number of channels = 3.

3-D patch that extends through the retained spectral bands; in the LiDAR data, it yields a 2-D slice with the same spatial extent. The resulting HSI and SAR/LiDAR patches are denoted as $X_h^{pca} \in \mathbb{R}^{s\times s\times C_h}$ and $X_{s/l}^{pca} \in \mathbb{R}^{s\times s\times C_{s/l}}$, where s is the patch size. In particular, for the SAR/LiDAR patch, the postextraction tensor formatting depends on its band count $C_{s/l}$. If $C_{s/l} > 1$, the patch is first permuted into the layout expected by the network. If $C_{s/l} = 1$, an explicit channel axis is unsqueezed.

B. Direction-Oriented Sparse Modeling

To jointly model heterogeneous structures and address the obstacles of vanilla 3-D convolution, we propose a DOSM module consisting of two parallel modality-specific convolutional streams. Each stream is tailored to the spatial–spectral or spatial–structural properties, with the corresponding direction-oriented sparse convolutions. In this way, neighboring samples that lie along the predefined axial sampling directions and are structurally representative are selected to perform discriminative and efficient intramodal representation learning while reducing computational overhead. As displayed in Fig. 4, the coordinate system is defined with the center voxel o at (0, 0, 0). The x-axis corresponds to the horizontal spatial dimension, the z-axis to the vertical spatial dimension, and the y-axis represents the channel dimension.

Two variants of direction-oriented sparse convolution designed based on different numbers of channels are illustrated in Fig. 5. We define a direction-oriented sparse kernel shape as mutually orthogonal planes intersecting at the central pixel. Denote p as the center pixel of a cubic neighborhood. Let k be the number of channels, and the set of spectral indices I can be given by

$$I = \begin{cases} \left\{-\frac{(k-1)}{2}, -\frac{(k-3)}{2}, \ldots, 0, \ldots, \frac{(k-3)}{2}, \frac{(k-1)}{2}\right\}, & \text{if } k \text{ is odd} \\ \left\{-\frac{k}{2}, -\frac{k}{2}+1, \ldots, \frac{k}{2}-1\right\}, & \text{if } k \text{ is even.} \end{cases} \tag{1}$$

As shown in Fig. 5(a), when the number of channels exceeds 3, a sparser directional pattern is applied to reduce computational burden and align with extended channel support. Instead of using full local neighborhoods, we extract voxels along the spectral axis with a fixed central cross section in spatial dimensions. The set of sampled positions $\mathcal{N}_m(p)$ is defined as

$$\mathcal{N}_m(p) = \{p + \delta \mid \delta \in \{(\pm 1, I, 0), (0, I, \pm 1), (0, I, 0)\}\}. \tag{2}$$

The number of sampled spectral neighbors depends on the size of k, and the sample set forms a sparse tube structure centered at p, preserving long-range dependency in the channel dimension while reducing spatial redundancy. When the number of channels equals 3, as displayed in Fig. 5(b), more voxels are introduced for balanced and complete spatial–spectral representation. The set of sampled voxels $\mathcal{N}_s(p)$ can be formulated as

$$\begin{aligned}\mathcal{N}_s(p) = \{p + \delta \mid \delta \in \{&(\pm 1, 0, 0), (\pm 1, 0, \pm 1), (\pm 1, \pm 1, 0) \\ &(0, \pm 1, \pm 1), (0, 0, \pm 1), (0, \pm 1, 0), (0, 0, 0)\}\}\end{aligned} \tag{3}$$

where only 19 voxels are retained, forming a cross-cube structure centered at p. These selected positions emphasize immediate semantic directions.

Let $p = (x_0, y_0, z_0)$ be the center pixel in the patch. For the HSI stream, a sparse neighborhood $\mathcal{N}_m(p)$ is first constructed for each center voxel p based on axial offsets along x-, y-, and z-directions. In each layer $l \in \{1, 2, 3\}$, the result $y_h^{(l)}(p)$ for the point p generated in the lth layer can be calculated as

$$y_h^{(l)}(p) = \sigma\left(\sum_{q\in\mathcal{N}_m(p)} w_h^{(l)}(q-p)\cdot x_h^{(l-1)}(q) + b_h^{(l)}\right) \tag{4}$$

where $q \in \mathcal{N}_m(p)$ indexes the sampled neighbors of the center voxel p, $x_h^{(l-1)}(q)$ is the feature at voxel q in the $(l-1)$th layer, $w_h^{(l)}(q-p)$ denotes the learnable weight masked by the modality-specific offset $(q-p)$, $b_h^{(l)}$ represents the bias term of the direction-oriented sparse convolution, and $\sigma(\cdot)$ is the LeakyReLU activation function. After the third convolution, the result is reshaped and followed by a standard 2-D convolution to obtain the high-level representation $y_h^{(4)}$ via the following calculation:

$$y_h^{(4)} = \sigma\left(w_h * Reshape\left(x_h^{(3)}\right) + b_h\right). \tag{5}$$

$Reshape(\cdot)$ is a reshape operation, w_h denotes the 2-D convolutional kernel weights, $x_h^{(3)}$ represents the output feature from the third direction-oriented sparse convolutional layer, and b_h is the bias term of 2-D convolution. Unlike existing sparse or anisotropic convolutions, DOSM treats sparsity as a structural prior. First, its kernel support is physically grounded: the spectral tube captures the long-range spectral continuity and local spatial redundancy characteristic of HSI, whereas the axial cross cube is tailored to the geometrically dense but spectrally sparse nature of SAR/LiDAR data. In contrast, dilated convolution employs isotropic dilation, while factorized 3-D convolutions remain dense and modality-agnostic.

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Second, this prior is deterministic and parameter-free, avoiding the additional parameters and potential instability associated with the learned offsets of deformable convolution under label-scarce conditions. Third, the two kernel variants shown in Fig. 5 make DOSM inherently modality-adaptive, allowing the modality-specific sampling structures to support intramodal mutual learning before intermodal integration.

For the SAR/LiDAR stream, we apply the 3-D convolutional kernel to capture dense spatial structure. Unlike HSI, SAR/LiDAR inputs provide rich spatial information but are spectrally sparse. Thus, the corresponding convolution $y_{s/l}^{(l)}(p)$ is computed as

$$y_{s/l}^{(l)}(p) = \sigma\left(\sum_{q\in\mathcal{N}_s(p)} w_{s/l}^{(l)}(q-p)\cdot x_{s/l}^{(l-1)}(q) + b_{s/l}^{(l)}\right) \tag{6}$$

where $q \in \mathcal{N}_s(p)$ indexes the sampled neighbors of the center voxel p, $x_{s/l}^{(l-1)}(q)$ is the feature at voxel q in the $(l-1)$th layer, $w_{s/l}^{(l)}(q-p)$ denotes the learnable weight masked by the modality-specific offset $(q-p)$, and $b_{s/l}^{(l)}$ is the bias term. The tensor is subsequently reshaped before being passed to a standard 2-D convolutional layer. This process can be defined as

$$y_{s/l}^{(4)} = \sigma\left(w_{s/l} * Reshape\left(x_{s/l}^{(3)}\right) + b_{s/l}\right) \tag{7}$$

where $Reshape(\cdot)$ is the reshape operation, $w_{s/l}$ is the 2-D convolutional kernel weights, $x_{s/l}^{(3)}$ is the output from the third direction-oriented sparse convolutional layer, and $b_{s/l}$ denotes the bias term of 2-D convolution.

Through the above process, the proposed DOSM enables each modality to retain its unique geographic cues while also extracting beneficial directional intramodal features, reducing interference, and alleviating the computational burden.

C. Continuous Contrast-Consensus Integration

To enhance the intermodal representation of heterogeneous data sources and preserve complementary spatial–spectral–structural information, a C3I module is designed. The proposed architecture enables mid-level semantic alignment and cross-modal interaction by explicitly modeling discriminative and complementary representation cross-modal attributes.

In each layer $l \in \{1,2,3\}$, assume X_h^l be the intermediate feature map obtained from the HSI stream at layer l, and $X_{s/l}^l$ represents the middle output from the SAR/LiDAR stream at layer l. The mean over the depth of each intermediate features is calculated as follows:

$$\bar{X}_h^l = \frac{1}{D_h^l}\sum_{d_h=1}^{D_h^l} X_h^l[:,:,d_h,:,:] \tag{8}$$

$$\bar{X}_{s/l}^l = \frac{1}{D_{s/l}^l}\sum_{d_{s/l}=1}^{D_{s/l}^l} X_{s/l}^l\left[:,:,d_{s/l},:,:\right] \tag{9}$$

where $\bar{X}_h^l$ and $\bar{X}_{s/l}^l$ are the mean spectral response of the HSI stream and SAR/LiDAR stream, respectively. D_h^l and $D_{s/l}^l$ represent the number of slices along the depth dimension in the HSI stream and SAR/LiDAR stream at layer l. d_h and $d_{s/l}$ index the slices along the depth dimension of the input HSI and SAR/LiDAR stream. Then, two saliency descriptors are constructed by computing both the elementwise absolute difference and the summation of mean features

$$\Delta^l = \left|\bar{X}_h^l - \bar{X}_{s/l}^l\right| \tag{10}$$

$$\Sigma^l = \left|\bar{X}_h^l + \bar{X}_{s/l}^l\right| \tag{11}$$

where $|\cdot|$ denotes the absolute value operator. Δ^l is the contrastive feature, and Σ^l represents the consensus features. In detail, the contrast process encodes modality-specific differences by computing the absolute difference between HSI and SAR/LiDAR feature statistics. The above contrastive information highlights discriminative cues that may only appear in one modality (e.g., spectral patterns in HSI, structural information in SAR, and elevation properties in LiDAR), thus enabling the network to learn modality complementarity. Consensus representation Σ^l captures modality-shared structures by summing depth-averaged features across corresponding layers. This promotes modality consensus, reinforcing signals that are consistent across both modalities and contributing to shared semantic understanding. Because the contrast branch is a difference operation, it could in principle amplify per-modality noise; C3I mitigates this by mean-pooling each stream along the depth dimension before differencing, which smooths band-level noise, and by regularizing both branches with a shared absolute-value operator that stabilizes polarity.

Building on the DOSM and C3I modules, we add a Transformer-based encoder to capture global contextual dependencies. Its output is then fused with the multilevel HSI–SAR/LiDAR feature interactions. For the HSI stream, output features are projected into a sequence of tokens $\tilde{y}_h^{(4)}$. Then, a learnable positional embedding p_h and a learnable class token cls_h are added to get the input sequence $Z_h^{(0)}$ for Transformer

$$Z_h^{(0)} = \left[cls_h; \tilde{y}_h^{(4)} + p_h\right]. \tag{12}$$

The Transformer encoder applies H layers of self-attention and feed-forward blocks. At each layer $h \in \{1,\ldots,H\}$, the update is

$$attn(Q,K,V) = softmax\left(\frac{QK^\top}{\sqrt{d}}\right)V \tag{13}$$

$$a_i = attn\left(XW_i^Q, XW_i^K, XW_i^V\right) \tag{14}$$

$$MHSA(X) = \varphi(a_1,\ldots,a_h)W^O + b^O \tag{15}$$

$$f(X) = \theta(XW_1 + b_1)\cdot W_2 + b_2 \tag{16}$$

$$\hat{Z}_h^{(l)'} = MHSA\left(LN\left(Z_h^{(l-1)}\right)\right) + Z_h^{(l-1)} \tag{17}$$

$$\hat{Z}_h^{(l)} = f\left(LN\left(\hat{Z}_h^{(l)'}\right)\right) + {}_h^{(l)'} \tag{18}$$

$$g(X) = \phi^{(2)}\left(\phi^{(1)}(X)\right) \tag{19}$$

$$Z_h = g\left(\hat{Z}_h^{(l)}[:,0,:]\right) \tag{20}$$

where Q, K, and V are the query, key, and value. $softmax(s)$ is softmax function and d is the dimension of each attention head. $X \in \mathbb{R}^{N\times d}$ denotes input feature, where N is the number of tokens and d is the number of input feature channels. $attn(\cdot)$ represents the self-attention calculation, W_i^Q, W_i^K, and

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W_i^V are the query, key, and value projection matrices for the ith head, a_i is the result of the self-attention operation, and $\varphi(\cdot)$ is the concatenation operation. W^O and b^O are the learnable projection and bias terms, respectively, to synthesize information from these different attentional perspectives into a global representation. $f(\cdot)$ denotes the feed-forward function, and $\theta(\cdot)$ represents the Gaussian error linear unit (GELU) activation function. $W_1 \in \mathbb{R}^{d\times d_f}$ and $W_2 \in \mathbb{R}^{d_f\times d_{o1}}$ are the parameters of the first and the second layer, respectively, where d_f denotes the hidden feature dimension and d_{o1} is the number of output channels. $b_1 \in \mathbb{R}^{d_f}$ and $b_2 \in \mathbb{R}^{d_{o1}}$ are the bias terms for two layers, respectively. The output of (16) is then projected into features by (17)–(31), and the generated logits from these calculations serve as the input to the loss function (32). $LN(\cdot)$ is the layer normalization function, and $MHSA(\cdot)$ represents multihead self-attention (MHSA). $\hat{Z}_h^{(l)}[:,0,:]$ corresponds to the class token, which serves as the summary representation of the input sequence. $Z_h^{(l)}$ is the output of lth layer, and $\hat{Z}_h^{(l)'}$ is the result of MHSA calculation in HSI stream. A fully connected group $g(X)$ consisting of two fully connected layers $\phi_{\ulcorner\urcorner}^{(1)}$ and $\phi_{\ulcorner\urcorner}^{(2)}$ are utilized to process this token. Z_h is the output of the HSI stream. Dropout layers are employed in the above process to avoid overfitting.

Similarly, the output of SAR/LiDAR stream $Z_{s/l}$ can be denoted as

$$\hat{Z}_{s/l}^{(l)'} = MHSA\left(LN\left(Z_{s/l}^{(l-1)}\right)\right) + Z_{s/l}^{(l-1)} \tag{21}$$

$$\hat{Z}_{s/l}^{(l)} = f\left(LN\left(\hat{Z}_{s/l}^{(l)'}\right)\right) +_{s/l}^{(l)'} \tag{22}$$

$$Z_{s/l} = g\left(\hat{Z}_{s/l}^{(l)}[:,0,:]\right) \tag{23}$$

where $Z_{s/l}^{(l)}$ represents the output of the lth layer, and $\hat{Z}_{s/l}^{(l)'}$ is the result of the MHSA calculation in the SAR/LiDAR stream.

To effectively leverage multimodal information from HSI and SAR/LiDAR data, a fusion strategy is employed, integrating three sources of high-level features: modality-specific Transformer outputs, and the statistical contrast and consensus responses across corresponding feature maps. Specifically, given the contrastive (difference) and consensus (addition) features as Z_{ct} and Z_{cs}

$$Z_{ct} = \phi_{dif}\left(\varphi\left(\Delta^1, \Delta^2, \Delta^3\right)\right) \tag{24}$$

$$Z_{cs} = \phi_{add}\left(\varphi\left(\Sigma^1, \Sigma^2, \Sigma^3\right)\right) \tag{25}$$

where $\phi_{dif}(\cdot)$ and $\phi_{add}(\cdot)$ are the fully connected layers for contrastive features and consensus features, respectively. The fused output of the model is formulated as a weighted combination

$$Z_{out} = \phi_{out}\left(\lambda_1 Z_h + \lambda_2 Z_{s/l} + \lambda_3 Z_{cs} - \lambda_4 Z_{ct}\right) \tag{26}$$

where λ_1, λ_2, λ_3, and λ_4 ($\lambda_3 = \lambda_4$) are the balancing factors controlling the contribution of HSI features Z_h, SAR/LiDAR features $Z_{s/l}$, consensus representation Z_{cs}, and contrast representation Z_{ct}. Note that contrast and consensus representation indicate complementary but inherently coupled perspectives. Unequal weighting leads to imbalanced optimization, either overemphasizing agreement or amplifying domain-specific biases, which can degrade generalization. From this point of view, assigning them equal balancing factors ($\lambda_3 = \lambda_4$) helps to symmetrically integrate their contributions. The sensitivity of performance to these factors is analyzed in Section III-I. $\phi_{out}(\cdot)$ denotes the fully connected layer that maps the aggregated representation into a c-dimensional space, where c is the number of classes. Hence, the fused output $Z_{out}^i \in R^c$ for each sample i serves as the input to the GEETF head. Especially, the subtraction of contrastive features during fusion acts as a regulatory term to mitigate modality conflicts. It serves to suppress conflicting signals between modalities, ensuring that the final representation emphasizes consistent and reliable regions while still benefiting from earlier contrast modeling [49].

By incorporating both contrast and consensus features, the network benefits from robust alignment of shared patterns and preservation of modality-unique contributions. These statistics, together with Transformer-encoded global representations from both branches, are fused through a weighted aggregation strategy, ultimately enhancing the performance under complex cross-modal conditions.

D. GEETF Head

HSI classification presents unique challenges, including high spectral dimensionality, spectral redundancy, class imbalance, and subtle interclass variance. To address these challenges, we utilize a GEETF head as the classifier, enabling class discrimination and geometric consistency while mitigating representation collapse.

Let Z_{out}^i be the output of sample i generated by the network. The classifier constructs a fixed, nontrainable matrix M, and each column m_c of M represents a class prototype. Let c be the number of classes and let d_{o2} be the feature dimension produced by (26). In the GEETF head, the fully connected layer in (26) maps the fused representation into a c-dimensional space, so that all subsequent features, scores, and logits share the same dimension. An orthonormal matrix $P \in \mathbb{R}^{d_{o2}\times c}$, satisfying $P^\top P = I$, is first generated, which serves as the foundation for defining balanced class prototypes. The reduced QR decomposition is applied to a randomly initialized matrix $A \in \mathbb{R}^{d_{o2}\times c}$ to obtain P, which can be formalized as $A = PR$, where $R \in \mathbb{R}^{c\times c}$ is an upper triangular matrix and $I \in \mathbb{R}^{c\times c}$ is the identity matrix. c serves as the number of the prototype class. The above calculation ensures that the columns of P are mutually orthogonal, forming the basis for constructing the ETF classifier matrix

$$M = \sqrt{\frac{c}{(c-1)}} \cdot P \cdot \left(I - \frac{1}{c}\mathbf{1}\mathbf{1}^\top\right) \tag{27}$$

where $\mathbf{1} \in \mathbb{R}^c$ denotes the all-ones vector, so that $\mathbf{1}\mathbf{1}^\mathbf{T} \in \mathbb{R}^{c\times c}$ is the all-ones matrix. $(I - 1/c\mathbf{1}\mathbf{1}^\top)$ ensures zero-mean and equal-angle spacing among class vectors, satisfying the ETF properties. This decomposition enforces geometric equilibrium, which stabilizes class representation even in high-dimensional, imbalanced feature spaces commonly found in HSI classification. The matrix M ensures equiangularity by

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construction: the inner product between any two class vectors m_i and m_j satisfies the following equation:

$$m_i^\top m_j = \begin{cases} 1, & i = j \\ -\dfrac{1}{c-1}, & i \neq j. \end{cases} \tag{28}$$

The above property ensures that the angle between class prototypes is uniformly distributed, achieving a tight frame geometry. Such structure avoids dominance of any class direction and encourages balanced decision boundaries.

The feature vector Z^i_{out} undergoes batch normalization followed by L2 normalization, which is given as follows:

$$Z^i_{outBN} = BN\left(Z^i_{out}\right) \tag{29}$$

$$\tilde{Z}^i_{outL2} = \frac{Z^i_{outBN}}{\|Z^i_{outBN}\|_2} \tag{30}$$

where $\|\cdot\|_2$ is the L2 normalization, $BN(\cdot)$ denotes the batch normalization, and $\tilde{Z}^i_{outBN} \in \mathbb{R}^{d_{o3}}$ corresponds to the output feature of sample i after batch normalization. $\tilde{Z}^i_{outL2} \in \mathbb{R}^c$ is the output after L2 normalization. Since $d_{o3} = c$, the normalized feature $\tilde{Z}^i_{outL2}$ and the score vector below lie in the same c-dimensional space. Then, the output $s^i_{out} \in \mathbb{R}^c$ is computed as the dot product between the normalized feature $\tilde{Z}^i_{outL2}$ and each column in M, and the decoupled final output $O^i_{out} \in \mathbb{R}^c$ can be defined as

$$s^i_{out} = \tilde{Z}^i_{outL2} M, \quad O^i_{out} = \tilde{Z}^i_{outL2} - s^i_{out}. \tag{31}$$

In this way, both terms of the subtraction are c-dimensional, and O^i_{out} serves directly as the logit vector for the cross-entropy loss in (32). The subtractive formulation encourages the network to decouple redundant projections and focus on learning residual discriminative cues. The computational path from the fusion output to the loss is dimensionally traceable at every step: 1) the fusion in (26) produces $Z^i_{out} \in \mathbb{R}^c$; 2) batch normalization in (29) and L2 normalization in (30) preserve this dimension, yielding $Z^i_{outBN} \in \mathbb{R}^c$ and the unit-norm feature $\tilde{Z}^i_{outL2} \in \mathbb{R}^c$; 3) the fixed ETF prototype matrix $M \in \mathbb{R}^{c\times c}$ constructed in (27) projects the normalized feature onto the c class prototypes, giving the score vector $s^i_{out} \in \mathbb{R}^c$, whose jth entry is the inner product $\langle \tilde{Z}^i_{outL2}, m_j\rangle$; 4) the decoupled logits $O^i_{out} = \tilde{Z}^i_{outL2} - s^i_{out} \in \mathbb{R}^c$ follow from (31); and 5) O^i_{out} is fed as the logit vector into the softmax cross-entropy loss in (32).

The geometric constraint is imposed implicitly through the ETF prototypes; the learned representation $\tilde{Z}^i_{out}$ is projected onto the ETF head, and the network is trained with the cross-entropy loss

$$\mathcal{L}_{CE} = -log\frac{e^{[O^i_{out}]}_{y_i}}{\sum_{j=1}^{c} e^{[O^i_{out}]}_j}, \quad \mathcal{L} = \mathcal{L}_{CE}\left(O^i_{out}, y_i\right) \tag{32}$$

where $e^{[O^i_{out}]}_{y_i}$ is the exponential of the logit score corresponding to the ground-truth class y_i of sample i, $e^{[O^i_{out}]}_j$ means the corresponding term for class j of sample i, and c is the total number of classes. $\mathcal{L}_{CE}$ is cross-entropy loss, and $\mathcal{L}$ is the total loss. Therefore, the GEETF head injects structural regularity into the classifier via fixed geometric templates. It effectively decouples representation learning from class weight learning, stabilizes optimization, and enhances classwise angular margins, resulting in better generalization.

On the multimodal classification tasks of HSI and SAR/LiDAR, we discover that the mutual features are distributed both intramodally and intermodally. With unified mutual learning, the model can extract the intrinsic information within modalities and model spatial–spectral–structural dependencies between modalities. Moreover, the GEETF head imposes equiangular constraints on the feature space, ensuring discriminative and balanced class separation.

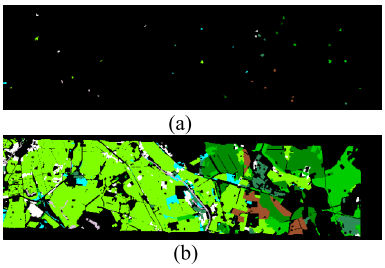


Fig. 6. Spatially disjoint (a) training and (b) test sets of Berlin dataset.

III. Experiments and Analysis

A. Dataset Description

Three real-world HSI scenarios and their corresponding SAR or LiDAR data with different modalities, resolutions, and sampling strategies are utilized in the experiments. The first dataset is HSI–SAR Augsburg, which is composed of 332 × 485 pixels. The HSI includes 180 spectral channels, while the SAR data has four features. The second one is HSI–LiDAR Houston2013, consisting of 349 × 1905 pixels. The HSI covers 144 spectral bands, and the LiDAR contains one band. The third dataset is HSI–SAR Berlin with 1723 × 476 pixels. The number of bands is 244 and 4 for HSI and SAR, respectively. The first and third datasets can be downloaded from https://github.com/danfenghong/ISPRS_S2FL, and the second dataset is available on https://hyperspectral.ee.uh.edu/?page_id=459 via data request.

To comprehensively evaluate the performance, both random sampling and spatially disjoint sampling are employed. For the Augsburg and Houston2013 datasets, 50 labeled samples per class are randomly drawn from the ground truth as the training set, and the remaining samples serve as the test set. For the Berlin dataset, the standard spatially disjoint training and test masks are adopted, as displayed in Fig. 6, with the numbers of training and test samples tabulated in Table I. Because neighboring pixels can be spatially correlated, the Berlin setting provides a complementary and stricter reference for assessing generalization. All results are averaged over ten runs with consecutive seeds 666–675, with the random state fixed per run by a deterministic seeding routine. For Augsburg and Houston2013, the training samples are redrawn under each run seed, whereas for Berlin the same spatially disjoint masks are used across runs. The pseudo-color images of HSI, SAR/LiDAR, and ground truth in the three datasets are illustrated in Figs. 7–9.

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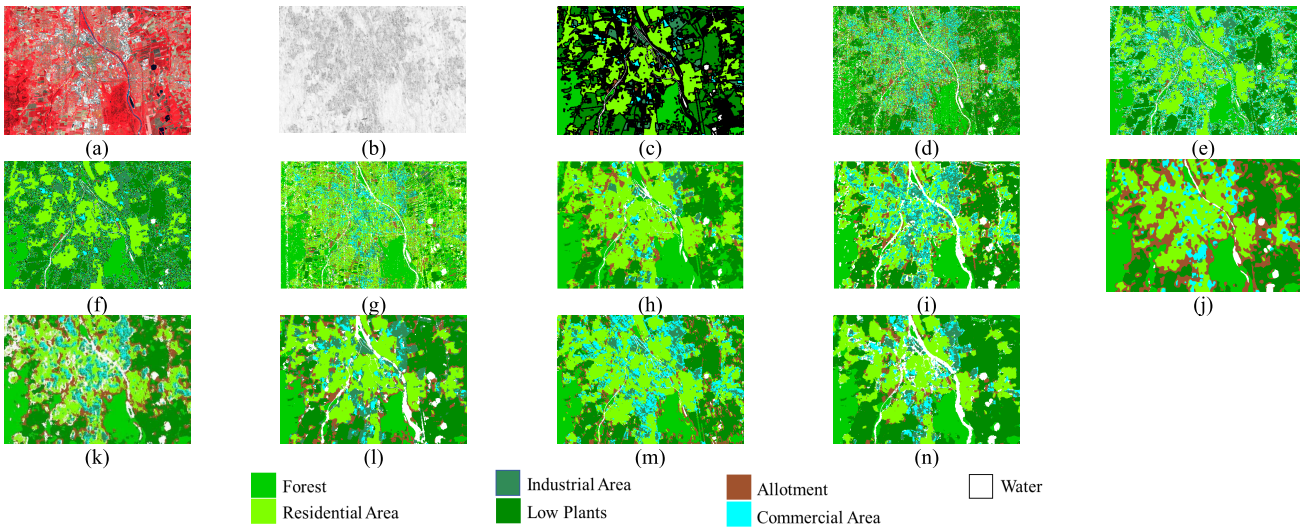


Fig. 7. Classification maps for the HSI–SAR Augsburg dataset. (a) Pseudo-color HSI. (b) SAR imagery. (c) Ground-truth map. (d) S2FL. (e) CCRNet. (f) AM3Net. (g) EndNet. (h) MAHiDFNet. (i) GLTNet. (j) DCMNet. (k) MSFMamba. (l) HSI. (m) SAR. (n) Proposed.

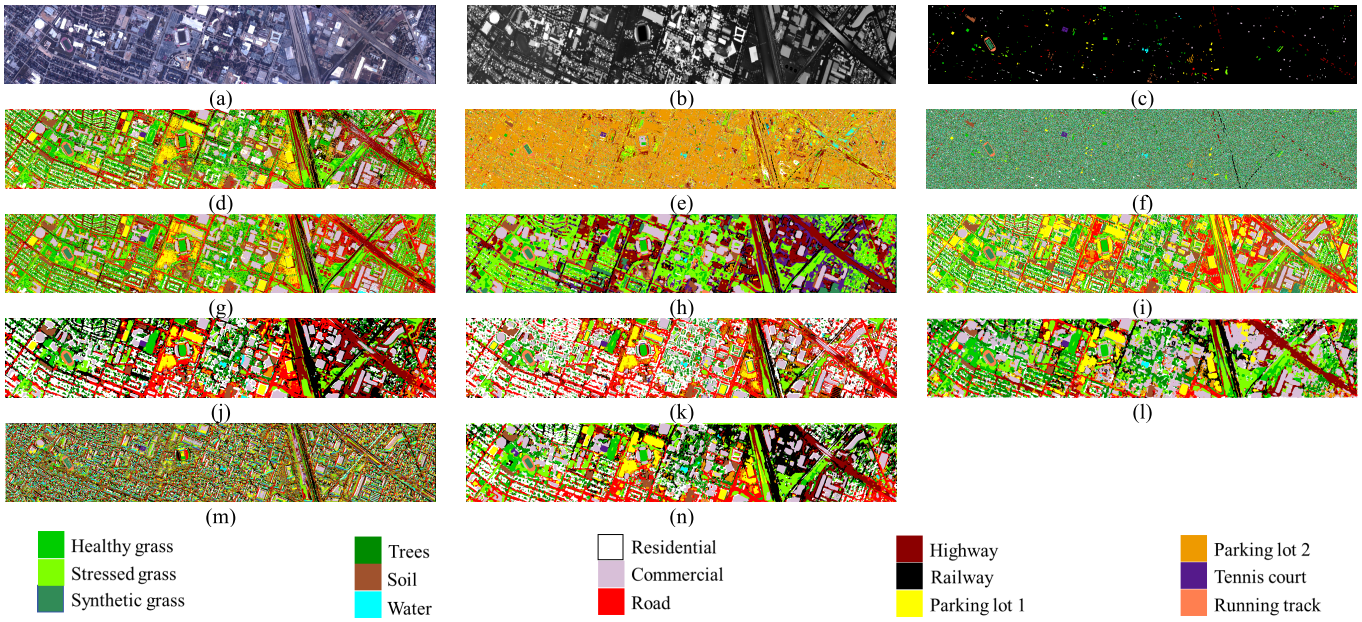


Fig. 8. Classification maps for HSI–LiDAR Houston2013 dataset. (a) Pseudo-color HSI. (b) LiDAR imagery. (c) Ground-truth map. (d) S2FL. (e) CCRNet. (f) AM3Net. (g) EndNet. (h) MAHiDFNet. (i) GLTNet. (j) DCMNet. (k) MSFMamba. (l) HSI. (m) LiDAR. (n) Proposed.

B. Experimental Setup

The proposed MERGE-Net is implemented within the PyTorch framework and executed on an NVIDIA GeForce RTX 3090Ti GPU. Adam optimizer is used with a learning rate of $1e^{-3}$ and a decay factor of $1e^{-3}$. For stable convergence, a multistep decay strategy is introduced to schedule the learning rate, reducing at half and five-sixths of the total epochs. The batch size is set to 256. For the Augsburg dataset, the model is trained for 200 epochs with a patch size of 15. On the Houston2013 scene, the epoch is 50, and the patch size is 15. For the Berlin data, the epoch and the patch size are set to 60 and 19, respectively.

To evaluate the effectiveness of the proposed MERGE-Net, we conduct extensive experiments by comparing it with state-of-the-art (SOTA) approaches, including shared and specific feature learning (S2FL) [50], cross-channel reconstruction network (CCRNet) [51], adaptive mutual-learning-based multimodal data fusion network (AM3Net) [52], deep encoder–decoder network (EndNet) [30], multi-attentive hierarchical dense fusion net (MAHiDFNet) [53], GLTNet [17], DCMNet [33], and multiscale feature fusion Mamba (MSFMamba) [54].

The performance of each algorithm is evaluated in overall accuracy (OA), Kappa coefficient (Kappa), and the corresponding standard deviation.

C. Performance Comparison

To comprehensively validate the performance, we compare the proposed MERGE-Net with advanced multimodal classification approaches as well as its own single-modal variants.

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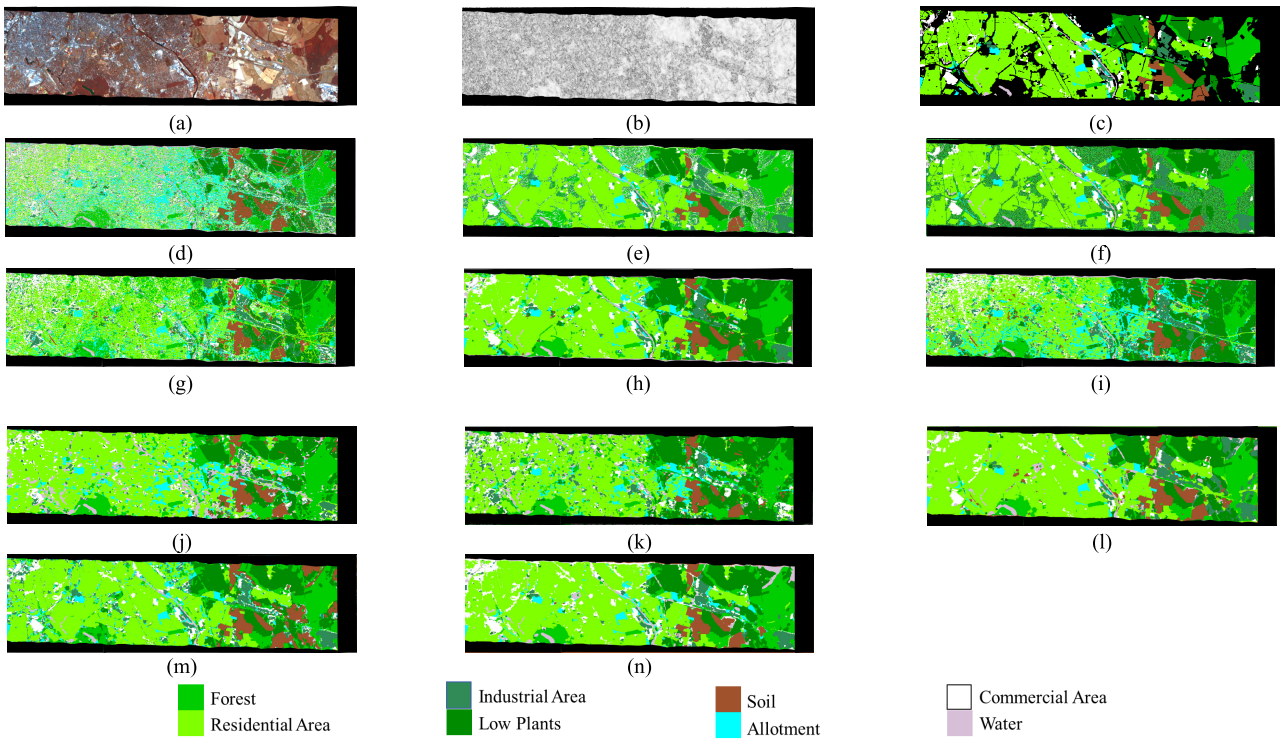


Fig. 9. Classification maps for the HSI–SAR Berlin dataset. (a) Pseudo-color HSI. (b) SAR imagery. (c) Ground-truth map. (d) S2FL. (e) CCRNet. (f) AM3Net. (g) EndNet. (h) MAHiDFNet. (i) GLTNet. (j) DCMNet. (k) MSFMamba. (l) HSI. (m) SAR. (n) Proposed.

TABLE I
TRAINING–TEST DISTRIBUTION OF BERLIN DATASET

Class No.	Class name	Training	Test
1	Forest	443	54511
2	Residential Area	423	268219
3	Industrial Area	499	19067
4	Low Plants	376	58906
5	Soil	331	17095
6	Allotment	280	13025
7	Commercial Area	298	24526
8	Water	170	6502
	Total	2820	461851

Tables II–IV list the quantitative performance among all the approaches. The best accuracy is bolded, and the second-best one is underlined. The proposed MERGE-Net attains the highest OA on all three benchmarks: HSI–SAR Augsburg (83.10%), HSI–LiDAR Houston2013 (92.98%), and HSI–SAR Berlin (71.35%), with corresponding Kappa values 82.43, 92.41, and 55.11. S2FL enhances multisource features but still trails MERGE-Net by >20% OA on Augsburg and >10% on Berlin. CCRNet's cross-channel relations help yet lack fine-grained intramodal preservation. AM3Net underperforms near mixed-material boundaries (OA $\geq$ 12% lower across datasets). EndNet is competitive on Houston2013 (0.49% OA behind) but less robust elsewhere. MAHiDFNet improves over most CNNs on Augsburg and Berlin yet is weak on Houston2013 (59.99% OA and 56.85 Kappa coefficient). GLTNet narrows the Augsburg gap to 2.13% OA, and DCMNet benefits from cross-modal interaction but is still 3.43% OA lower on Houston2013. The OA of MSFMamba is 2%–15% lower than MERGE-Net on Augsburg and Houston2013. MERGE-Net attains the highest OA on three datasets and the best OA–Kappa pair on Augsburg and Houston2013. On Berlin, MSFMamba achieves a slightly higher Kappa (57.68 versus 55.11). Across these datasets spanning two modality combinations and both random and spatially disjoint sampling, this consistent ranking indicates stable performance under the evaluated conditions.

Relative to multimodal approaches, MERGE-Net yields OA gains of 2.13%–32.81% and Kappa gains of 8.23–46.39 on Augsburg, 0.49%–32.99% OA and 0.54–35.56 Kappa on Houston2013, and 0.51%–28.79% OA on Berlin; for Berlin Kappa, MERGE-Net ranks second yet remains 2.23–28.76 higher than all multimodal methods except MSFMamba. Compared with the strongest monomodal baselines in each dataset, MERGE-Net improves OA by 48.41% (Augsburg versus SAR 34.69), 1.02% (Houston2013 versus HSI 91.96), and 16.79% (Berlin versus HSI 54.56). Multimodal fusion is generally expected to outperform single-modal variants by aggregating complementary information. However, the results show that monomodal variants can surpass some multimodal approaches. This occurs when complementary properties are not fully exploited. Approaches using shallow or simple fusion, without intermodal and intramodal modeling, are easily dominated by the lower quality modality. In addition, neglecting directional features and using poor optimization amplifies noise and further degrades accuracy.

Qualitative results (see Figs. 7–9) show that MERGE-Net yields the cleanest maps: markedly reduced salt-and-pepper noise, sharper and more coherent boundaries, and better

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TABLE II
QUANTITATIVE CLASSIFICATION RESULTS OF THE HSI–SAR AUGSBURG DATASET

Class No.	S2FL	CCRNet	AM3Net	EndNet	MAHiDFNet	GLTNet	DCMNet	MSFMamba	HSI	SAR	Proposed
1	88.83	88.49	77.64	83.84	52.17	92.96	87.39	90.37	40.56	50.71	93.01
	(3.43)	(2.18)	(3.87)	(9.21)	(7.26)	(6.30)	(21.69)	(0.89)	(33.05)	(42.44)	(2.25)
2	37.69	56.06	36.13	67.57	82.12	75.15	89.47	66.23	24.97	36.44	79.93
	(3.24)	(3.07)	(11.90)	(4.48)	(9.50)	(7.92)	(11.86)	(1.68)	(30.28)	(18.78)	(7.23)
3	46.59	47.86	59.80	36.37	72.62	55.00	56.35	42.64	39.26	34.18	54.96
	(3.47)	(5.84)	(7.98)	(5.61)	(6.17)	(18.53)	(11.85)	(3.61)	(27.36)	(12.52)	(14.27)
4	80.30	53.34	49.89	16.68	91.16	88.43	68.87	66.98	32.73	24.98	89.10
	(1.05)	(8.25)	(8.24)	(10.03)	(6.30)	(6.44)	(20.36)	(3.04)	(31.42)	(25.85)	(4.78)
5	80.65	78.93	71.37	60.84	6.42	90.29	65.71	78.66	58.29	36.08	78.63
	(3.13)	(4.11)	(8.62)	(21.02)	(2.22)	(12.12)	(38.52)	(2.48)	(31.71)	(13.97)	(5.34)
6	41.93	41.44	59.15	43.26	27.22	37.32	22.17	44.46	28.24	44.84	50.26
	(4.81)	(6.80)	(7.95)	(5.51)	(3.27)	(22.94)	(14.34)	(5.05)	(14.99)	(15.60)	(11.83)
7	60.14	49.92	59.26	63.15	51.46	66.31	39.72	53.77	54.85	18.89	58.61
	(1.01)	(3.45)	(4.95)	(11.06)	(11.49)	(6.09)	(5.19)	(2.48)	(10.40)	(19.54)	(8.09)
OA	62.41	60.07	50.29	50.73	64.24	80.97	77.93	68.93	31.89	34.69	**83.10**
(%)	(1.16)	(2.92)	(2.67)	(2.82)	(4.94)	(3.65)	(7.02)	(0.98)	(22.06)	(17.67)	**(3.24)**
Kappa	52.54	48.52	38.41	36.04	52.00	74.20	68.45	59.76	21.67	24.93	**82.43**
×100	(1.24)	(2.87)	(2.25)	(4.78)	(6.69)	(4.67)	(11.71)	(1.17)	(23.22)	(16.42)	**(3.06)**

TABLE III
QUANTITATIVE CLASSIFICATION RESULTS OF THE HSI–LIDAR HOUSTON2013 DATASET

Class No.	S2FL	CCRNet	AM3Net	EndNet	MAHiDFNet	GLTNet	DCMNet	MSFMamba	HSI	LiDAR	Proposed
1	96.10	95.19	78.09	96.19	72.53	92.30	88.54	88.75	90.54	8.48	92.41
	(1.19)	(0.70)	(5.04)	(1.74)	(11.11)	(4.99)	(6.95)	(1.56)	(3.65)	(7.29)	(2.91)
2	97.49	96.50	66.36	97.41	50.01	81.22	92.43	96.32	97.56	18.04	97.66
	(1.45)	(1.19)	(6.96)	(2.23)	(7.00)	(32.36)	(5.40)	(0.82)	(1.87)	(5.02)	(2.11)
3	100.00	99.63	93.74	99.41	51.48	99.56	99.97	99.81	99.69	19.29	99.04
	(0.00)	(0.45)	(0.96)	(0.66)	(10.24)	(0.48)	(0.07)	(0.18)	(0.33)	(7.25)	(1.41)
4	98.01	96.58	69.34	96.60	100.00	90.44	96.42	89.66	94.04	49.85	89.03
	(0.70)	(1.66)	(10.19)	(1.05)	(0.00)	(9.29)	(0.98)	(1.47)	(3.25)	(8.89)	(7.46)
5	99.30	98.02	90.82	98.62	98.53	59.85	99.98	97.65	99.48	12.47	98.76
	(0.33)	(0.80)	(2.96)	(0.61)	(1.17)	(54.64)	(0.04)	(0.38)	(0.49)	(4.25)	(1.26)
6	99.64	91.85	94.84	97.38	99.66	90.79	97.67	98.25	98.18	26.76	97.53
	(0.26)	(3.12)	(2.57)	(1.57)	(0.75)	(4.59)	(3.07)	(0.90)	(2.57)	(7.62)	(1.96)
7	89.64	89.57	75.38	90.85	100.00	76.58	89.26	91.16	81.07	39.05	92.64
	(2.02)	(3.81)	(6.72)	(3.23)	(0.00)	(8.37)	(4.55)	(1.56)	(9.51)	(3.97)	(3.43)
8	87.84	90.89	78.58	93.05	91.52	80.55	95.26	92.71	82.83	43.05	90.69
	(3.08)	(1.45)	(2.07)	(2.39)	(7.77)	(12.94)	(1.31)	(1.92)	(5.99)	(8.43)	(4.87)
9	76.64	80.78	62.40	82.61	98.71	45.10	64.03	79.76	81.58	21.11	85.62
	(2.26)	(4.68)	(7.83)	(2.66)	(0.97)	(46.02)	(12.43)	(2.14)	(7.29)	(5.79)	(2.16)
10	90.60	91.59	91.66	92.42	30.73	21.20	72.18	92.03	93.47	26.56	92.15
	(3.16)	(3.17)	(3.32)	(2.51)	(2.54)	(35.02)	(7.63)	(2.36)	(3.38)	(5.10)	(5.83)
11	88.51	90.53	92.76	92.88	77.60	10.01	97.43	93.24	93.42	37.99	96.57
	(2.56)	(2.90)	(2.40)	(2.32)	(9.31)	(22.38)	(1.90)	(1.52)	(2.35)	(10.49)	(1.98)
12	85.44	85.93	80.86	80.27	49.51	25.89	82.47	75.85	93.42	14.40	87.07
	(1.72)	(3.97)	(3.47)	(9.27)	(43.97)	(19.19)	(3.69)	(1.86)	(2.66)	(1.83)	(5.95)
13	68.26	81.19	91.38	73.56	98.68	66.75	93.22	87.06	92.84	35.04	92.60
	(1.46)	(8.53)	(1.35)	(7.98)	(1.05)	(21.31)	(1.77)	(0.92)	(3.12)	(9.68)	(4.24)
14	99.52	97.09	99.88	99.31	49.50	34.74	100.00	100.00	99.95	24.55	100.00
	(0.22)	(2.80)	(0.15)	(0.52)	(4.50)	(47.91)	(0.00)	(0.00)	(0.12)	(5.88)	(0.00)
15	99.64	97.90	86.25	99.34	93.92	99.66	100.00	99.96	99.84	9.93	99.25
	(0.39)	(1.83)	(4.28)	(0.53)	(1.43)	(0.59)	(0.00)	(0.07)	(0.16)	(5.55)	(1.16)
OA (%)	91.47	92.04	80.69	92.49	59.99	62.46	89.55	90.93	91.96	26.18	**92.98**
	(0.64)	(0.52)	(1.40)	(0.39)	(2.36)	(2.78)	(0.30)	(0.42)	(1.20)	(2.56)	**(1.32)**
Kappa×100	90.77	91.39	79.09	91.87	56.85	59.41	88.70	90.18	91.30	20.65	**92.41**
	(0.70)	(0.56)	(1.50)	(0.42)	(2.53)	(3.05)	(0.32)	(0.46)	(1.29)	(2.71)	**(1.43)**

preservation of fine-scale structures and urban edges. By comparison, S2FL exhibits coarse boundaries and class mixing, consistent with insufficient multisource feature extraction and missing contrast cues. CCRNet suppresses some noise via cross-modal interaction but still over-smooths and misclassifies small land-cover elements, most visibly on Houston2013. AM3Net better retains small objects yet remains error-prone in the complex Houston2013 scene, indicating suboptimal contextual consistency. EndNet shows boundary distortions and omission of small classes, suggesting over-compression of high-frequency content (e.g., roads and residential areas). MAHiDFNet integrates multimodal cues more effectively,

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TABLE IV
QUANTITATIVE CLASSIFICATION RESULTS OF THE HSI–SAR BERLIN DATASET

Class No.	S2FL	CCRNet	AM3Net	EndNet	MAHiDFNet	GLTNet	DCMNet	MSFMamba	HSI	SAR	Proposed
1	80.15	66.15	56.45	75.50	60.89	61.60	76.27	73.76	71.20	51.55	73.31
	(0.95)	(5.93)	(4.51)	(3.37)	(9.82)	(13.11)	(28.18)	(2.73)	(19.76)	(3.82)	(3.79)
2	51.72	53.92	44.91	61.82	92.28	63.84	70.63	73.57	52.53	26.99	82.52
	(1.89)	(6.56)	(11.13)	(2.52)	(1.85)	(2.56)	(15.84)	(0.76)	(19.19)	(3.07)	(3.26)
3	47.05	45.89	33.02	51.51	47.68	43.79	41.03	40.72	39.82	41.26	30.97
	(1.68)	(4.69)	(6.51)	(5.01)	(6.17)	(11.77)	(9.55)	(3.87)	(11.61)	(1.44)	(13.17)
4	66.28	76.78	27.24	53.11	64.40	83.03	60.86	86.24	52.04	52.85	69.22
	(0.79)	(2.22)	(8.19)	(7.69)	(6.05)	(11.38)	(22.27)	(1.47)	(14.19)	(4.78)	(17.69)
5	82.74	81.90	58.48	69.72	57.10	81.97	72.18	76.30	72.18	62.89	58.48
	(0.17)	(2.86)	(10.71)	(3.04)	(1.87)	(7.80)	(40.86)	(2.64)	(22.34)	(5.88)	(33.09)
6	58.62	71.34	40.53	55.48	16.75	66.28	75.33	65.83	58.62	18.03	17.70
	(1.02)	(1.58)	(13.99)	(10.46)	(3.23)	(9.93)	(15.42)	(4.00)	(26.80)	(1.19)	(16.37)
7	25.66	41.80	25.32	33.38	33.40	38.42	18.98	23.90	51.43	20.38	30.61
	(1.02)	(4.24)	(3.95)	(3.05)	(4.84)	(9.64)	(9.27)	(2.20)	(13.03)	(1.92)	(14.69)
8	55.01	66.94	23.02	61.17	49.66	68.14	73.56	54.88	22.02	30.37	26.78
	(0.83)	(3.70)	(7.06)	(1.58)	(9.63)	(5.44)	(16.79)	(2.53)	(3.40)	(4.91)	(27.60)
OA (%)	56.75	59.01	42.56	60.50	<u>66.70</u>	64.65	66.28	70.84	54.56	34.55	**71.35**
	(1.08)	(3.26)	(5.65)	(1.87)	<u>(0.68)</u>	(1.83)	(1.66)	(0.68)	(10.11)	(1.77)	**(0.59)**
Kappa×100	42.64	46.11	26.35	44.83	52.88	51.24	49.12	**57.68**	40.93	21.65	<u>55.11</u>
	(0.90)	(2.82)	(2.75)	(1.76)	(0.43)	(1.98)	(8.77)	**(0.89)**	(7.76)	(1.52)	<u>(1.65)</u>

but residual oversmoothing persists, implying incomplete modeling of intermodal/intramodal dependencies. GLTNet leverages global–local interactions to preserve large homogeneous regions (notably in Berlin) while still confusing spectrally similar classes in dense urban zones. DCMNet produces visually continuous maps with lower noise on broad land covers, but its tendency to over-smooth narrow features leads to omission errors (e.g., highways and commercial strips). MSFMamba struggles to segment challenging commercial areas, and very few regions are identified. Overall, MERGE-Net provides more spatial details with less noise across scenes.

MERGE-Net includes two monomodal variants (HSI and SAR/LiDAR). The HSI variant yields more coherent vegetation shapes but confuses complex built-up areas without complementary structure. The SAR/LiDAR variant captures geometry (e.g., railways or highways) yet lacks spectral discrimination, causing systematic confusions among spectrally similar covers (e.g., healthy and stressed grass). By contrast, the multimodal MERGE-Net improves both coherence and boundary fidelity: in HSI–SAR Augsburg it separates low plants from residential areas while preserving narrow boundaries; in HSI–LiDAR Houston2013 it keeps small, intricate objects (running tracks and roads) with sharp edges and reduces salt-and-pepper noise in homogeneous regions; in HSI–SAR Berlin it maintains large vegetation blocks and correctly identifies small nonvegetation inclusions. These gains arise from: DOSM, which preserves orientation-specific, multiview dependencies within each modality to enhance intrinsic structure and mitigate cross-modal misalignment; C3I, which extracts hierarchical contrast and consensus cues to strengthen cross-modal representation; and GEETF, which enforces compact, well-separated prototypes to stabilize decision boundaries in ambiguous classes.

The quantitative and qualitative comparisons show a similar trend, demonstrating the capability between fine-grained detail preservation and large-scale consistency.

TABLE V
EXPERIMENTAL RESULTS OF ABLATION STUDY

Method		Backbone	+ DOSM	+ C3I	+ GEETF
Backbone		√	√	√	√
DOSM			√	√	√
C3I				√	√
GEETF					√
Augsburg	OA (%)	78.73 ±4.36	78.82 ±3.63	82.27 ±0.56	**83.10 ±3.24**
	Kappa ×100	60.89 ±11.29	78.98 ±3.55	79.76 ±3.19	**82.43 ±3.06**
Houston 2013	OA (%)	89.00 ±5.97	91.38 ±1.80	92.81 ±0.74	**92.98 ±1.32**
	Kappa ×100	88.10 ±6.45	90.68 ±1.94	92.22 ±0.81	**92.41 ±1.43**
Berlin	OA (%)	52.64 ±29.20	61.31 ±16.78	62.37 ±16.03	**71.35 ±0.59**
	Kappa ×100	41.29 ±24.36	44.60 ±11.20	46.62 ±16.14	**55.11 ±1.65**

D. Ablation Study

To evaluate the effectiveness of each architectural component, we conduct ablation experiments on three diverse multisource remote sensing datasets.

As visualized in Table V (the best outcome is bolded), each component demonstrates consistent improvements in both OA and Kappa coefficient on all datasets. Take the HSI–SAR Augsburg dataset as an an example, the backbone achieves an OA of 78.73%, which marginally improves to 78.82% with DOSM. However, the introduction of C3I results in a significant gain to 82.27%, while the full configuration with GEETF reaches the highest OA of 83.10%, yielding a cumulative improvement of 4.37%. In terms of the Kappa coefficient, it rises from 60.89 to 82.43 through successive module integration. In the HSI–LiDAR Houston2013 dataset, a similar trend is observed: the backbone's OA of 89.00% increases to 92.81% with DOSM and C3I and peaks at 92.98% when GEETF is employed. Notably, the HSI–SAR Berlin

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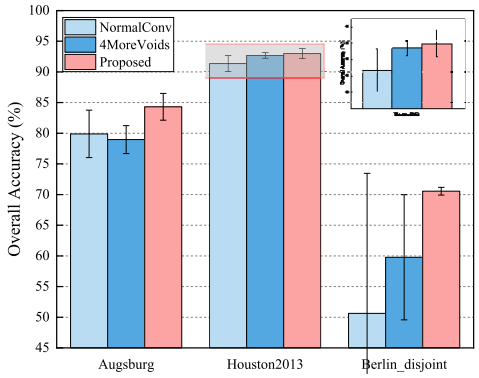


Fig. 10. OA of different convolutional designs.

dataset, which is more imbalanced and noisier as a result of disjoint training samples, shows even more substantial improvements. The OA jumps from 52.64% (backbone) to 71.35% (increased by 18.71%) with all proposed architectures, and the Kappa increases from 41.29 to 55.11 (+13.82), demonstrating the performance of the proposed framework under challenging conditions. In addition, when all modules are added, the standard deviation is also reduced and competitive, reflecting the robustness of the proposed approach.

Overall, DOSM facilitates intramodal encoding by capturing intrinsic information with directional sensitivity; C3I strengthens intermodal interaction by reinforcing mutual dependencies across heterogeneous modalities; and GEETF promotes geometric alignment and improves class separability. Their contributions are complementary but not uniform: DOSM and C3I provide the largest gains on Augsburg, DOSM contributes most on Houston2013, while GEETF and DOSM provide the largest gains on the spatially disjoint Berlin scenario.

E. Analysis on Convolutional Design

To assess the effect of the designed DOSM, we conduct a comparative analysis against standard convolution (NormalConv) and a sparser convolution structure (4MoreVoids), which introduces four additional void positions into the convolution kernel.

As shown in Fig. 10, on the HSI–SAR Augsburg dataset, the proposed approach achieves an OA of 84.31% $\pm$ 2.18%, outperforming both NormalConv (79.89% $\pm$ 3.86%) and 4MoreVoids (78.96% $\pm$ 2.28%), corresponding to gains of 4.42 and 5.35 percentage points, respectively. The direction-aware design likely reduces intraclass ambiguity and better preserves geometric patterns, yielding more stable predictions with lower variance. On the Houston2013 dataset, the proposed approach achieves 92.99% $\pm$ 0.81%, marginally higher than 4MoreVoids (92.70% $\pm$ 0.44%) and NormalConv (91.36% $\pm$ 1.30%). Interestingly, 4MoreVoids, despite its sparser structure, achieves higher accuracy than NormalConv on this dataset. We attribute this to the benefits of sparsity, which helps suppress local redundancy and overfitting in dense data regions. However, because 4MoreVoids lacks directional modeling, its performance remains below that of the proposed approach. On the most challenging Berlin dataset with the disjoint sampling strategy, the proposed approach shows the largest gain, reaching 70.55% $\pm$ 0.63%, substantially ahead of 4MoreVoids (59.78% $\pm$ 10.21%) and NormalConv (50.63% $\pm$ 22.83%), i.e., improvements of 19.92 and 10.77 percentage points, respectively. The dramatic reduction in standard deviation (from 22.83 to 0.63) further highlights the robustness of the proposed operator in sparse and irregular urban landscapes.

Accordingly, the proposed DOSM explicitly encodes spatial–spectral–structural patterns, enabling localized precision and, on the three datasets tested, more stable performance, notably the reduced standard deviation on the disjoint-sampled Berlin set.

F. Effect of Integration Patterns

Where and how many layers to integrate is an important question in the proposed C3I approach. To explore this question, a comprehensive study by activating contrast and consensus operations at different stages is conducted. For example, C3Ii refers to consensus and contrast modeling in the *i*th layer, with C3Iij extending this to the *i*th and *j*th layers, and C3Iijk applying it to all layers. A key challenge is the tradeoff between contrast and consensus. Focusing only on contrast emphasizes differences but may lack semantic alignment, leading to instability. Conversely, an exclusive focus on consensus merges shared information, which can suppress modality-specific cues and cause insufficient feature extraction. For these reasons, contrast and consensus always coexist in these variants.

Fig. 11 compares seven C3I variants across all datasets. On HSI–SAR Augsburg, C3I123 attains the highest metrics (OA >84% and Kappa >78%), clearly outperforming single-layer (C3I1/2/3) and two-layer (C3I12/13/23) settings; the jump from C3I13 to C3I123 (~3% OA) highlights the value of intermediate-layer fusion for multiscale context. On HSI–LiDAR Houston2013, where other methods are already strong, C3I123 still yields a consistent margin (OA $\approx$ 93% and Kappa >92%), sharpening decision boundaries and enriching multigranular features, while C3I13/23 fluctuate due to weaker layer interaction. The largest gaps appear on HSI–SAR Berlin (spatially disjoint sampling): single-layer C3I2/3 degrade markedly (OA < 60% and Kappa < 50%), whereas C3I123 achieves OA >70% with Kappa $\approx$ 55% and markedly tighter error bars. Although multilayer fusion generally improves contextual modeling, two-layer integration can underperform single-layer fusion when mutual learning is imbalanced: fusing only two adjacent or noncomplementary stages can amplify redundancy or noise and break representational continuity, yielding inconsistent features and suboptimal boundaries (e.g., C3I12 missing high-level semantics, C3I23 losing fine-grained edges, and C3I13 omitting mid-level context). In contrast, full three-layer integration (C3I123) preserves hierarchical complementarity and yields more stable performance across the three datasets.

Therefore, the full three-layer integration (C3I123) ensures a consistent and complete encoding of both contrastive and consensual patterns across the network hierarchy, leading to superior and more reliable performance.

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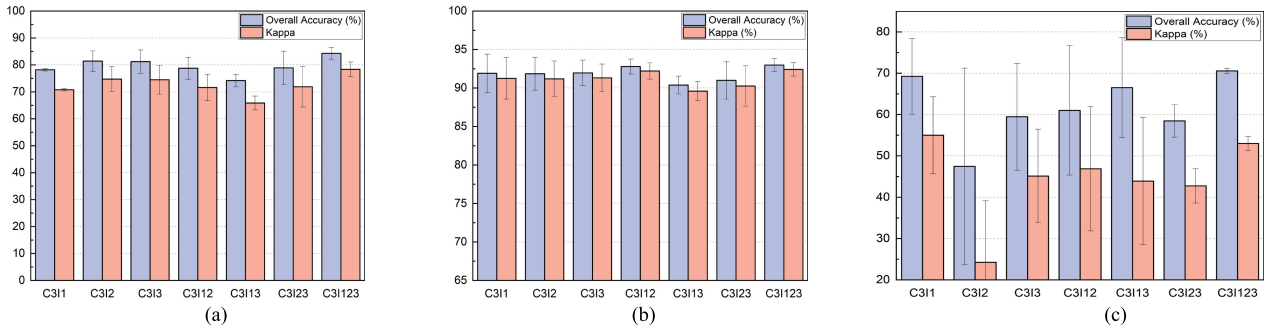


Fig. 11. Performance of diverse integration patterns. (a) Augsburg dataset. (b) Houston2013 dataset. (c) Berlin dataset.

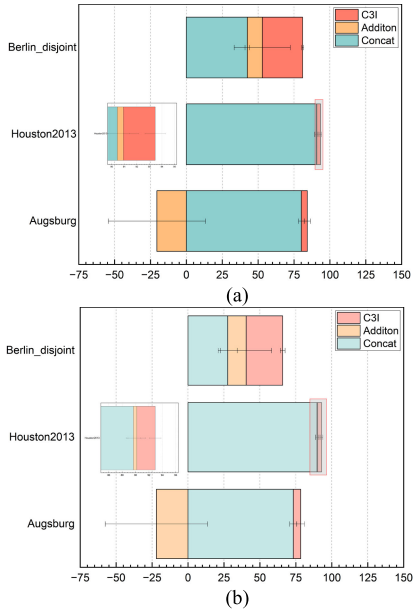


Fig. 12. Increments of (a) OA and (b) Kappa coefficient.

G. Influence of Fusion Strategies

Beyond the questions of where and how many times to integrate, the strategy of how to fuse is another critical issue that remains to be explored. There are two mainstream fusion strategies in most of the existing work, including elementwise addition and feature concatenation (abbreviated as Concat in Fig. 12). Fig. 12 presents the increments based on OA and the Kappa coefficient under the above strategies and the proposed C3I.

It can be observed that C3I yields consistent and dataset-wide gains over both addition and concatenation. In OA [see Fig. 12(a)], C3I improves Berlin by more than 28%, clearly surpassing existing two strategies. It also leads Houston2013 by >2% even where concatenation is already strong, and raises Augsburg by ≈ 4%. The same pattern holds for Kappa [see Fig. 12(b)]: C3I exceeds both alternatives by >25% on Berlin and >5% on Augsburg, while addition suffers a large drop (≈ 22%) on Augsburg and yields < 13% improvement on Berlin. These results reflect the limitations of the other approaches: addition's elementwise fusion ignores contrastive structure across layers, promoting redundancy and adverse mixing. Concatenation inflates dimensionality without modeling intermodal relations, increasing computation and redundancy, hindering gradient flow, and offering limited accuracy gains relative to its cost.

In contrast, the proposed C3I strategy explicitly models contrast and consensus information across multiple semantic levels, enabling more effective fusion. The results highlight its capability to preserve complementary features while avoiding the pitfalls of over-simplified (addition) or over-parameterized (concatenation) designs.

TABLE VI
ASSESSMENT OF VARIED ABSOLUTE VALUE CRITERION

Method		No_Abs	Abs_Ct	Abs_Cs	Proposed
Augsburg	OA (%)	48.32 ±19.23	49.18 ±17.37	47.48 ±20.06	**83.10 ±3.24**
	Kappa × 100	44.23 ±11.32	28.80 ±18.40	21.94 ±8.55	**82.43 ±3.06**
Houston 2013	OA (%)	92.76 ±0.66	91.93 ±1.38	91.72 ±1.85	**92.98 ±1.32**
	Kappa × 100	92.17 ±0.72	91.27 ±1.49	91.04 ±2.01	**92.41 ±1.43**
Berlin	OA (%)	57.12 ±13.73	41.11 ±26.54	30.69 ±11.44	**71.35 ±0.59**
	Kappa × 100	36.62 ±23.62	36.03 ±23.98	35.75 ±24.56	**55.11 ±1.65**

H. Role of Absolute Value Criterion

To investigate the role of distinct absolute value criteria on unified mutual learning, we evaluate four configurations: no absolute value (No_Abs), applying absolute value only to the contrast features (Abs_Ct), only to the consensus features (Abs_Cs), and the proposed strategy that jointly regulates both.

As summarized in Table VI (best in bold and second best underlined), the proposed criterion attains the top performance on all three datasets, reaching 83.10% OA and 82.43 Kappa coefficient on Augsburg, 92.98% and 92.41 on Houston2013, and 71.35% and 55.11 on Berlin. Relative to No_Abs, gains are especially pronounced on Augsburg (+34.78% OA) and Berlin (+14.23% OA). Against variants that apply absolute values to only one branch (Abs_Ct or Abs_Cs), our criterion improves OA by up to about 40% across datasets. Notably, partial regularization can underperform the unregularized baseline: on Berlin, Abs_Ct = 41.11% and Abs_Cs = 30.69% fall well below No_Abs = 57.12%, with a similar pattern on

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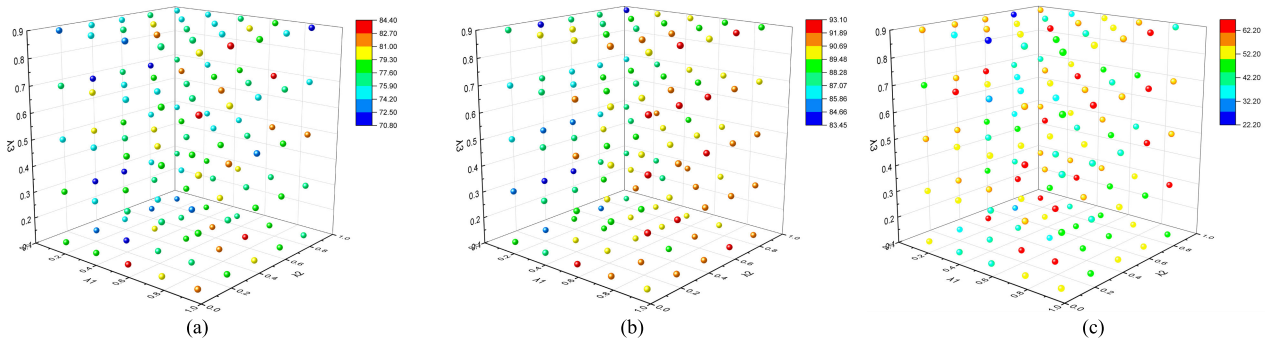


Fig. 13. Result with diverse balancing factors. (a) Augsburg dataset. (b) Houston2013 dataset. (c) Berlin dataset.

TABLE VII
PARAMETER COUNTS, INFERENCE MEMORY, PER-SAMPLE LATENCY, AND OA OF DIFFERENT APPROACHES

Approach	Params (M)	Inference Memory (MB)	Latency (ms)	OA (%)
CCRNet	**0.055**	**10 / 18 / 18**	**0.54**	60.07 / 92.04 / 59.01
AM3Net	2.78	106 / 165 / 165	1.2	50.29 / 80.69 / 42.56
EndNet	0.088	18	0.61	50.73 / 92.49 / 60.50
MAHiDFNet	122.17	540.58	2.77	64.24 / 59.99 / 66.70
GLTNet	0.40	53 / 75 / 75	7.3	80.97 / 62.46 / 64.65
DCMNet	2.20	35	4.6	77.93 / 89.55 / 66.28
MSFMamba	1.03 / 1.24 / 3.24	106 / 167 / 167	3.2 / 3.2 / 6.4	68.93 / 90.93 / 70.84
MERGE-Net (ours)	89.10 / 80.93 / 187.58	554 / 530 / 1126	4.1	**83.10 / 92.98 / 71.35**

Houston2013. We attribute this to asymmetric distortion from one-sided absolute-value operations, which disrupts feature distributions, induces inconsistent gradients, and misaligns contrast and consensus streams during backpropagation. While No_Abs preserves natural polarity and better maintains complementarity, it remains vulnerable to sign conflicts; jointly regularizing both contrast and consensus in C3I stabilizes polarity and mitigates sign ambiguity.

Thus, by applying the absolute value operation to both components simultaneously, the proposed criterion can establish a more robust and intelligible feature interaction, resulting in consistently enhanced performance on all datasets.

I. Discussion of Balancing Factors

Balancing factors serve to quantify the contributions of various feature representations. To explore the optimal setting of balancing factors in (26), 125 unique λ-combination experiments are performed on each dataset. The following analysis characterizes the sensitivity of performance to the balancing factors.

As displayed in Fig. 13, for the Augsburg dataset, the highest OA on the λ grid, 84.38%, occurs at $\lambda_1 = 0.9$, $\lambda_2 = 0.3$, and $\lambda_3 = \lambda_4 = 0.9$. This configuration emphasizes the dominant role of HSI features in this dataset while preserving strong guidance from contrast-consensus terms. The relatively low weight of SAR information suggests that this modality contributes supplementary but not primary discriminative cues on the Augsburg dataset. In the case of the Houston2013 dataset, the grid maximum (OA of 93.09%) occurs at $\lambda_1 = 0.9$, $\lambda_2 = 0.7$, and $\lambda_3 = \lambda_4 = 0.9$. The above change indicates that a higher weight to LiDAR features leads to a notable performance gain. This highlights the modality's complementary value in capturing elevation and geometric cues that are less evident in HSI data alone. Furthermore, the contrast-consensus component is still essential for enhancing multimodal integration. On the difficult Berlin dataset, the highest OA on the grid is observed at $\lambda_1 = 0.9$, $\lambda_2 = 0.7$, and $\lambda_3 = \lambda_4 = 0.5$. It can be seen that both HSI and SAR information are crucial and beneficial. Moreover, a carefully chosen balancing factor for contrast-consensus representation still contributes positively to the overall performance. This result can demonstrate that even in challenging scenarios, the integration of contrastive and consensus features facilitates more effective and robust unified mutual learning.

The aforementioned results show that performance is relatively stable over a broad range of balancing factors, although dataset-specific tuning yields a further gain.

J. Evaluation of Computational Efficiency

Besides classification accuracy, computational efficiency is another factor to be considered for multimodal joint classification. Fig. 14 illustrates the OA and time cost for representative approaches and MERGE-Net. In addition, Table VII lists the parameter count (Params), inference memory, per-sample latency (Latency), and OA of all parametric methods under the same implementation setting. The best performance is bolded. Slash-separated values are given in the order Augsburg/Houston2013/Berlin; OA is reported per dataset for all approaches, while Params, inference memory, and Latency are shown per dataset where they vary and otherwise as a single dataset-invariant value.

As shown in Table VII, MERGE-Net attains the highest OA on all datasets at a per-sample latency of 4.1 ms, comparable to DCMNet (4.6 ms) and MSFMamba (3.2–6.4 ms) and below GLTNet (7.3 ms). Lightweight approaches such as CCRNet, AM3Net, and EndNet achieve sub-millisecond to 1.2-ms inference but markedly lower accuracy, below 61% on the

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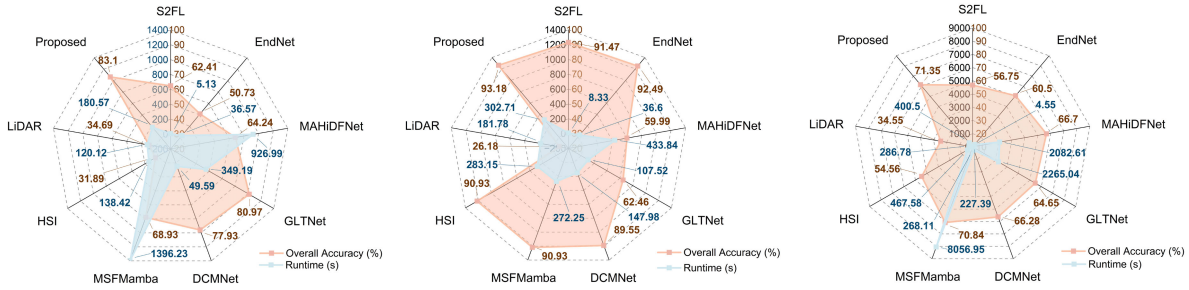


Fig. 14. OA and runtime of various approaches. (a) Augsburg dataset. (b) Houston2013 dataset. (c) Berlin dataset.

Augsburg dataset. Consistent with this, Fig. 14 shows that on the Augsburg dataset MSFMamba and MAHiDFNet incur the highest overall time cost (1396.23 and 926.99 s, respectively) without achieving the best accuracy, and GLTNet, the second-best method in OA (80.97%, about 2% below MERGE-Net), requires roughly twice the time of MERGE-Net due to its multiple Transformer-based encoders. While MERGE-Net is among the largest models in parameter count and inference memory, this cost is dominated by the fully connected layer that flattens the feature map before classification, whose input dimension scales with each dataset's patch size and channel count; the DOSM, C3I, and GEETF modules contribute a small fraction of the total. MERGE-Net thus delivers the best accuracy while keeping latency competitive.

IV. Conclusion

To effectively model both intramodal and intermodal representations and promote discriminative yet balanced class separation, MERGE-Net is proposed for joint classification of HSI and SAR/LiDAR data. The framework comprises three key components: 1) a DOSM module, which enhances direction-aware intrinsic information within individual modalities; 2) a C3I structure, which refines spatial–spectral–structural representations across modalities through multilevel and intermodal interactions; and 3) a GEETF head, which facilitates the formation of compact, stable, and discriminative features for robust prediction. Extensive experiments on three widely used datasets under both random and spatially disjoint sampling validate the effectiveness of the proposed approach in harmonizing modality-specific features with shared semantic cues. MERGE-Net achieves advanced performance while maintaining competitive computational efficiency. As future work, incorporating dynamic modality selection and uncertainty-aware learning presents a promising direction to further enhance adaptability in heterogeneous sensing environments. A fully learned directional operator that adapts the sampling geometry per location also merits further exploration. In addition, extending the spatially disjoint protocol to additional datasets could offer a more conservative assessment of generalization and is a useful direction.

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